# Homework Assignment 2
# Vectors and Tensors

All problems are graded on effort only.

**Lecture Notes:** Please complete all starred and double-starred exercises from section R.1-4

**Practice Problems:** The remaining problems are practice multiple-choice problems, such as will appear on the miderm and final exams. Give them a try to test your grasp of the lecture notes.

**Problem 1:** Using our tensor notation with indices referring to a basis of a vector space $V$, the quantities denoted by ${T_{\alpha\beta}}^{\gamma}$ are

(A) scalars

(B) components of a (2,1) tensor

(C) components of a multilinear scalar-valued function from $V \times V \times V^*$

(D) all of the above

**Problem 2:** For an inner product space, if vectors $u$ and $w$ have corresponding metric duals $\tilde{u}$ and $\tilde{w}$, then $\tilde{u}(w)$ equals:

(A) $(w|u)$ (B) $(u|w) + \overline{(u|w)}$ (C) $\tilde{w}(u)$ (D) $\overline{\tilde{w}(u)}$

where the bar denotes complex conjugation.